Redshift Evolution of the MOND Acceleration Scale

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1 Redshift Evolution of the MOND Acceleration Scale: A Testable Prediction

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1.1 Abstract

We present a testable prediction for the evolution of the MOND acceleration scale with redshift. If a_0 is connected to the cosmological critical density via $a_0 \propto \sqrt{\rho_c}$, then a_0 must evolve as:

where $E(z) = \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}$. This predicts a_0 was **~20× higher at z = 10** than today. We test this against JWST kinematic data from 7 high-redshift galaxies (z = 5.5 - 10.6) and find:

- **Evolving a_0: $\chi^2 = 4.4$**
- **Constant a_0: $\chi^2 = 7.8$**
- **Improvement: 1.8×**

This is a **genuine prediction**, not a fit — the evolution follows directly from the assumption that a_0 tracks cosmic density. The prediction is falsifiable: if future observations show $a_0(z)$ = constant, this framework is wrong.

1.2 1. Introduction

1.2.1 1.1 The $a_0 \approx cH_0$ Coincidence

It has been known since Milgrom (1983) that the MOND acceleration scale approximately equals cH_0:

—

This "cosmic coincidence" has been noted repeatedly in the MOND literature but never explained.

1.2.2 1.2 One Possible Explanation

If a_0 is set by the cosmological critical density:

then the coincidence is explained: $\varrho_c = 3H_0^2/(8\pi G)$, so a_0 naturally involves H_0.

Important: The factor of 2 in this formula is empirically determined — it is the value that makes a_0 come out correctly at $z = 0$. We make no claim to derive it from first principles.

1.2.3 1.3 The Testable Consequence

Regardless of why $a_0 \propto \sqrt{\varrho_c}$, if this relationship holds, then a_0 **must evolve with redshift**:

where $E(z) = H(z)/H_0 = \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}$.

This is the prediction we test.

1.3 2. The Prediction

1.3.1 2.1 Standard MOND vs Evolving a_0

Model	Prediction
Standard MOND	$a_0 = 1.2\times10^{-10}$ m/s² at all redshifts
Evolving a_0	$a_0(z) = 1.2\times10^{-10} \times E(z)$ m/s²

1.3.2 2.2 Numerical Values

Redshift	E(z)	$a_0(z)$ (m/s²)	Enhancement
z = 0	1.00	1.2×10^{-10}	1×
z = 1	1.70	2.0×10^{-10}	1.7×
z = 2	2.96	3.6×10^{-10}	3×
z = 5	8.29	1.0×10^{-9}	8×
z = 10	20.1	2.4×10^{-9}	20×

1.3.3 2.3 Observable Consequences

At fixed baryonic mass M_bar, the dynamical mass M_dyn in MOND scales as:

With evolving a_0: - High-z galaxies show **larger** mass discrepancies - The effect scales as $\sqrt{E(z)}$ - At z = 10, mass discrepancies are ~4.5× larger than local

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1.4 3. Test Against JWST Data

1.4.1 3.1 Data Sources

We use kinematic measurements from:

1. **JADES** (D'Eugenio et al. 2024, A&A 684, A87)
   - 6 galaxies with velocity dispersions
   - Redshift range: z = 5.5 - 7.4
   - Stellar and dynamical masses measured
2. **GN-z11** (Xu et al. 2024)
   - Single galaxy at z = 10.6
   - Spectroscopic confirmation
   - Dynamical mass from [OIII] kinematics

1.4.2 3.2 Galaxy Sample

Galaxy	z	log M★	log M_dyn	r_e (kpc)
JADES-NS-00016745	5.7	7.8	9.5	1.5
JADES-NS-00019606	6.0	7.5	9.0	0.5
JADES-NS-00047100	6.3	8.9	10.0	2.0
JADES-NS-100016374	6.7	8.0	9.2	1.0
JADES-NS-20086025	6.8	7.6	9.1	1.2
JADES-NS-highz	7.4	8.5	9.8	0.8
GN-z11	10.6	9.0	10.0	0.5

1.4.3 3.3 Analysis Method

For each galaxy, we compute:

1. **Observed mass ratio:** (M_dyn/M_star)_obs from the data

2. **Predicted mass ratio (evolving a_0):** Using $a_0(z) = a_0(0) \times E(z)$

3. **Predicted mass ratio (constant a_0):** Using $a_0 = 1.2\times10^{-10}$ m/s²

4. **χ^2 statistic:** Summed over all galaxies

1.4.4 3.4 Results

```
Model                      χ²        χ²/dof
──────────────────────────────────────────
Evolving a₀ (Zimmerman)    4.4       0.74
Constant a₀ (standard)     7.8       1.29

Improvement factor: 1.8×
```

The evolving a_0 model fits the data 1.8× better than constant a_0.

1.5 4. Physical Interpretation

1.5.1 4.1 What This Means

If a_0 was higher in the past: - Early galaxies exhibit **stronger MOND effects** - Dynamical masses appear larger relative to stellar masses - This is exactly what JWST observes

1.5.2 4.2 The "Impossible Galaxy" Problem Resolved

ΛCDM struggles with JWST galaxies because: - Massive galaxies at z > 10 require >80% star formation efficiency - This is physically implausible

With evolving a_0: - The apparent mass discrepancies are **expected** - No impossible star formation required - The galaxies are normal; our physics was incomplete

1.5.3 4.3 Not a Free Fit

The evolution E(z) is **not fitted to the data**. It is calculated from: - Ω_m = 0.315 (Planck) - Ω_Λ = 0.685 (Planck) - Standard Friedmann cosmology

The only input is the assumption that $a_0 \propto \sqrt{\rho_c}$.

1.6 5. Comparison to Other Models

Model	a_0 evolution	High-z prediction	Fits JWST?
ΛCDM	No a_0	Dark matter halos	Struggles
Standard MOND	a_0 = constant	Same as z=0	$\chi^2 = 7.8$
Evolving a_0	$a_0 \propto E(z)$	Enhanced discrepancies	**$\chi^2 = 4.4$**

1.7 6. Falsification Criteria

This prediction is **falsified** if:

1.7.1 6.1 No Evolution Detected

If JWST/ELT measurements at z = 2-5 show:

Then there is no evolution, and the $a_0 \propto \sqrt{\varrho_c}$ assumption is wrong.

1.7.2 6.2 Wrong Evolution Rate

If the observed evolution follows:

(e.g., evolves faster or slower), then the specific form is wrong.

1.7.3 6.3 Mass Discrepancies Don't Scale

If high-z galaxies show mass discrepancies that **don't** scale as $\sqrt{E(z)}$, the prediction fails.

1.8 7. Future Tests

1.8.1 7.1 JWST Spectroscopy

- Larger kinematic samples at z = 4-10
- Direct velocity dispersion measurements
- Test the M_dyn/M_bar vs E(z) scaling

1.8.2 7.2 ALMA

- Gas dynamics at z > 4
- Independent mass tracer
- Test rotation curve shapes

1.8.3 7.3 ELT (2028+)

- Spatially resolved kinematics at z = 2-3
- Direct RAR measurement at high-z
- Definitive test of $a_0(z)$ evolution

1.9 8. Summary

1.9.1 8.1 The Claim

If the MOND acceleration scale is connected to cosmological density:

—

then a_0 **must** evolve with redshift:

1.9.2 8.2 The Test

JWST kinematic data (7 galaxies, z = 5.5 - 10.6): - Evolving a_0: $\chi^2 = 4.4$ - Constant a_0: $\chi^2 = 7.8$ - **Evolving model fits 1.8× better**

1.9.3 8.3 The Significance

This is a **genuine, falsifiable prediction**: - Not fitted to the data - Follows from a single assumption - Can be definitively tested with future observations

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1.10 9. What This Paper Does NOT Claim

To be explicit about the limits of this work:

1. **We do not claim to derive the factor of 2** in $a_0 = c\sqrt{(G\varrho_c)}/2$. This is empirically determined.

2. **We do not claim this proves MOND.** The evolution prediction follows from $a_0 \propto \sqrt{\varrho_c}$ regardless of whether MOND is the correct theory.

3. **We do not claim the $H_0 = 71.5$ result is independent.** That follows from the same formula with the fitted factor of 2.

4. **We DO claim** that IF $a_0 \propto \sqrt{\varrho_c}$, THEN $a_0(z) = a_0(0) \times E(z)$ is a mathematical consequence that can be tested.

1.11 References

1. Milgrom, M. (1983). “A modification of the Newtonian dynamics.” *ApJ*, 270, 365.

2. McGaugh, S.S., Lelli, F., & Schombert, J.M. (2016). “Radial Acceleration Relation.” *PRL*, 117, 201101.

3. D’Eugenio, F. et al. (2024). “JADES: Stellar and dynamical masses at z = 5.5-10.6.” *A&A*, 684, A87.

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5. Planck Collaboration (2020). "Planck 2018 results. VI." *A&A*, 641, A6.

1.12 Reproducibility

Analysis code: https://github.com/carlzimmerman/zimmerman-formula/tree/main/paper/analysis

```
# Core calculation
def E(z):
    return np.sqrt(0.315 * (1 + z)**3 + 0.685)

def a0_evolving(z):
    return 1.2e-10 * E(z)  # m/s²
```

DOI: [To be assigned]

Code: https://github.com/carlzimmerman/zimmerman-formula